Abel Lawrence Peirson

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Education

Stanford University — *Ph.D. in Physics* *2017 - Mar 2023*
- Thesis: Relativistic Jets in the Era of X-ray Polarimetry.
- GPA: 4.0 (in required coursework).

University of Oxford — *MPhys in Physics* *2013 - 2017*
- First Class Honours with Distinction – top 10% of graduating class.
- College: Christ Church

Experience

Citadel GQS — New York / Miami *2023 - Present*
Quantitative Researcher
- Alpha research: sequence modelling and structured deep learning.

Google Brain — Mountain View, CA *June - Dec 2022*
Student Researcher
- Optimization for deep learning, NTK-NNPK correspondence.
- Mentors: Rohan Anil / Ehsan Amid.

G-Research — London, UK *June - Sep 2021*
Quantitative Researcher Intern
- Algorithmic trading detection.

Peirson & Freedman — Stanford, CA *Aug 2018 - 2019*
Co-founder
- Built iOS app ***Dank Learning*** that uses RNNs to generate memes. [Webpage]

Honors & Awards

Stanford Data Science Scholar ($100k+) — Stanford, CA *2021*
Future Investigator in NASA Earth and Space Science and Technology FINESST ($160k+) — Stanford, CA *2019*
Hooke & Roach Prizes: top STEM student, Christ Church — Oxford, UK *2017*
Oxford International Strategy Scholarship (CERN) — Oxford, UK *2016*
Gold, British Physics Olympiad — London, UK *2013*

Featured in: NASA Blog (2022) — Hogg's Research (2022) — NVIDIA AI Podcast, ep. 68 (2018) — TechCrunch (2018)

Open Source (★1000+)

Python — C/C++ — JAX — PyTorch — TensorFlow

MLCommons Algorithmic Efficiency: Benchmark and competition for neural network training. [Code][Webpage]

ixpeobssim: Simulation and analysis framework for the Imaging X-ray Polarimetry Explorer. [Code][Webpage]

MulensModel: Python package for modelling gravitational microlensing events. [Code][Paper][Webpage]

SSCpol: Polarized relativistic jet simulation in C with Python wrapper. [Code][Paper]

Dank Learning: 'Show and Tell' image captioning for meme generation in TensorFlow. [Code][Paper][Webpage]

Selected Publications (h-index 44)

— Machine Learning —

Benchmarking Neural Network Training Algorithms [Code]
G.E.Dahl, et al. incl. **A.L.Peirson** 2306.07179, 2023

Fishy: Layerwise Fisher Approximation for Higher-Order Neural Network Optimization
A.L.Peirson, E.Amid, Y.Chen, V.Feinberg, M.K.Warmuth, R.Anil *NeurIPS HITY workshop*, 2022

Neural Network Analysis of X-ray Polarimeter Data
A.L.Peirson, *The Handbook of X-ray and Gamma Ray Astrophysics*, *Springer Nature*, 2022

A Deep Ensemble Approach to X-ray Polarimetry
A.L.Peirson, R.W.Romani *NeurIPS ML4PS workshop*, 2021

Towards Optimal Signal Extraction for Imaging X-ray Polarimetry
A.L.Peirson, R.W.Romani. *ApJ*, 920, 40, 2021

Deep Ensemble Analysis for Imaging X-ray Polarimetry
A.L.Peirson, R.W.Romani, H.L.Marshall, J.F.Steiner, L.Baldini. *NIMA*, 986, 2021

Dank Learning: Generating Memes Using Deep Neural Networks [Code][TechCrunch][The Next Web]
A.L.Peirson & E.M.Tolunay, 1806.04510, 2018

— Astrophysics —

X-ray Polarization of BL Lacertae in Outburst
A.L.Peirson, M.Negro, I.Liodakis et al. *ApJ Letters*, 948, L25, 2023

Polarized Blazar X-rays imply Particle Acceleration in Shocks
I.Liodakis, et al. (IXPE collaboration, incl. **A.L.Peirson**) *Nature*, 611, 677, 2022

Polarized X-rays Constrain the Disk-Jet Geometry in the Black Hole X-ray Binary Cygnus X-1
H.Krawczynski, et al. (IXPE collaboration, incl. **A.L.Peirson**) *Science*, 378, 650, 2022

Testing High-Energy Emission Models for Blazars with X-ray Polarimetry
A.L.Peirson, I.Liodakis, R.W.Romani *ApJ*, 931, 59, 2022

New Tests of Millilensing in the Blazar PKS 1413+135 [Code]
A.L.Peirson, I.Liodakis, A.C.S.Readhead et al. *ApJ*, 927, 24, 2022

The Polarization Behavior of Relativistic Synchrotron Self-Compton Jets [Code]
A.L.Peirson, R.W.Romani. *ApJ*, 885, 76, 2019

Prospects for Detecting X-ray Polarization in Blazar Jets
I.Liodakis, **A.L.Peirson**, R.W.Romani. *ApJ*, 880, 29, 2019

The Polarization Behavior of Relativistic Synchrotron Jets
A.L.Peirson, R.W.Romani. *ApJ*, 864, 140, 2018